

Worksheet 1B: Fractions in the Wild — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A: Physics

1. (a) 0; $\gamma = 1$ (b) 0.25; 0.75 (c) 1 (d) 0 (e) $\gamma \rightarrow \infty$
2. (a) Equal (b) Approaches 0 (c) Slower
3. (a) $\text{rest_mass} \times c^2$ (b) $\rightarrow \infty$ (c) Would need infinite energy
4. (a) $\text{mass} \times v$ (classical momentum) (b) $\rightarrow \infty$

Part B: Probability

5. (a) Small (b) Gets larger (c) Small numerator / large denominator = small result
6. (a) Much larger numerator (b) More meaningful for common conditions
7. (a) Ratio increases (b) Ratio decreases

Part C: Information Theory

8. (a) Close to 1 (b) Low surprise (c) Very large (d) High surprise (e) More surprising
9. (a) Biased coin (b) Biased coin (c) Biased coin; fewer bits needed for predictable outcomes
10. (a) Close to 0 (b) 1 (c) Yes; means file got larger (bad compression)

Part D: Machine Learning

11. (a) 1; 0.5 (b) $\rightarrow 0$; approaches 1 (c) $\rightarrow \infty$; approaches 0 (d) No
12. (a) Valid probability (b) Neutral input = uncertain output
13. (a) They sum to 1 (b) Approaches 1 (c) All equal (1/3 each if three items)
14. (a) `initial_rate` (b) 0 (c) Big steps early to explore, small steps later to fine-tune
15. (a) Decrease (b) Big step (c) Limits step size
16. (a) 0 (b) 1 (c) Division by near-zero; normalized values explode

Part E: Exponential

- 17. (a) 2 (b) 2.25 (c) ≈ 2.613 (d) ≈ 2.714 (e) Compounding every instant
- 18. (a) $1/2$ (b) $1/4$ (c) $1/1024 \approx 0.001$ (d) No, asymptotic
- 19. (a) 0 (b) 0.5 (c) 0.9 (d) 0.99 (e) 1

Part F: Classification

- 20. (a) 0.9 (b) Down (c) 1
- 21. (a) 0.9 (b) 0; recall = 1 (c) Plummets (many false positives)
- 22. (a) 1 (b) 0 (c) 0.6 (d) Harmonic mean punishes imbalance more
- 23. (a) 99% (b) Zero (c) High accuracy despite useless predictions

Part G: Series

- 24. (a) 2 (b) $3/2 = 1.5$ (c) 3 (d) Series diverges; doesn't converge
- 25. (a) 6 (b) 100 (c) ∞

Part H: Challenge

- 26. (a) $\rightarrow 0$ (b) $\rightarrow 1$ (c) $\rightarrow K$ (carrying capacity) (d) Substitute and simplify
- 27. (a) 0; 1 (b) Larger (c) More negative (d) Approaches 0
- 28. (a) ≈ 0.9999 (b) ≈ 0.00005 each (c) Model "focuses" on highest-scored item

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